

Algorithms for Large Graphs

COMPSCI 753

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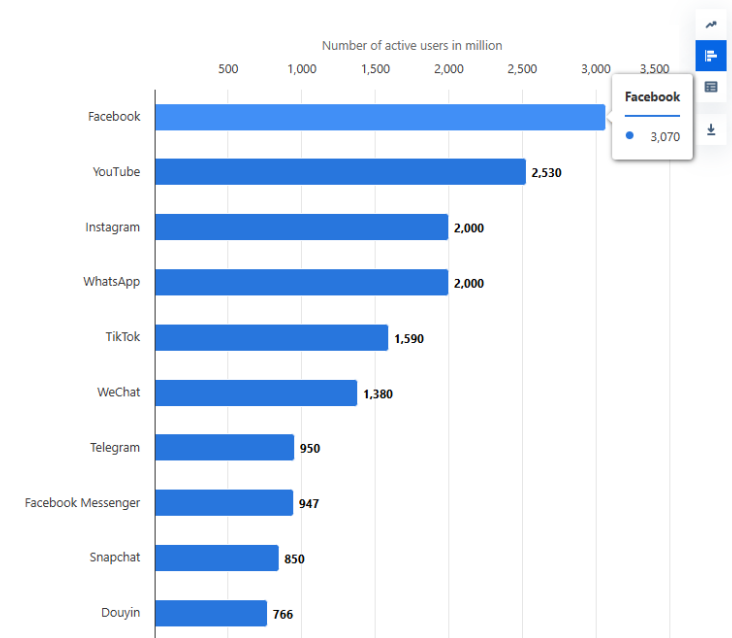
Slides adapted from “Mining of Massive Datasets” <http://mmds.org>

Graph data

- Social networks

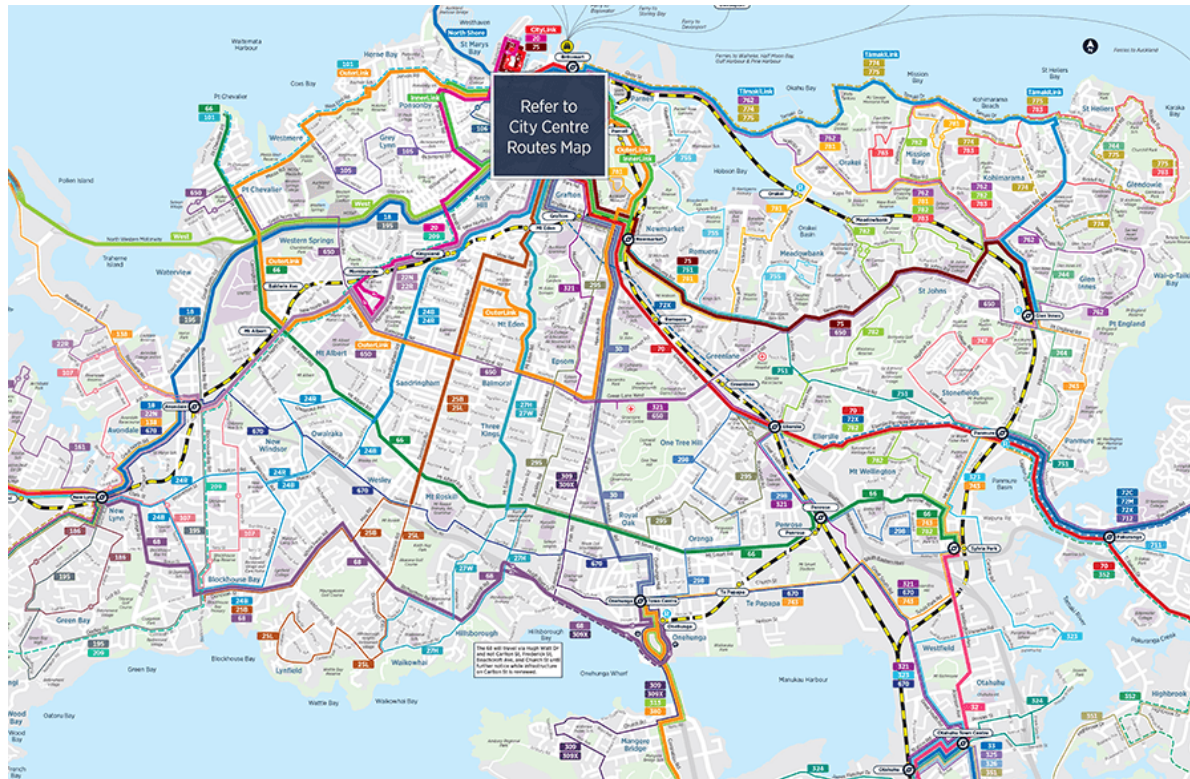


Monthly active users (up to February 2025, in millions, from statista.com)



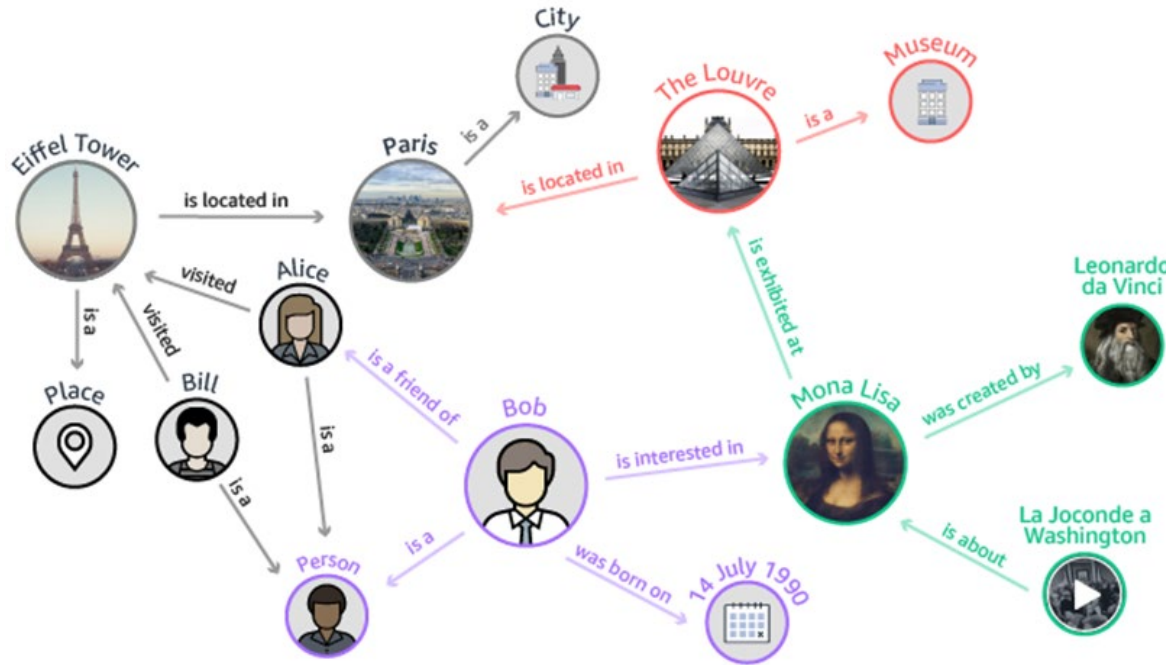
Graph data

- Transportation networks



Graph data

- Knowledge graphs



Yago

10M nodes

120M facts

DBPedia

4.58M nodes

3B facts

ConceptNet

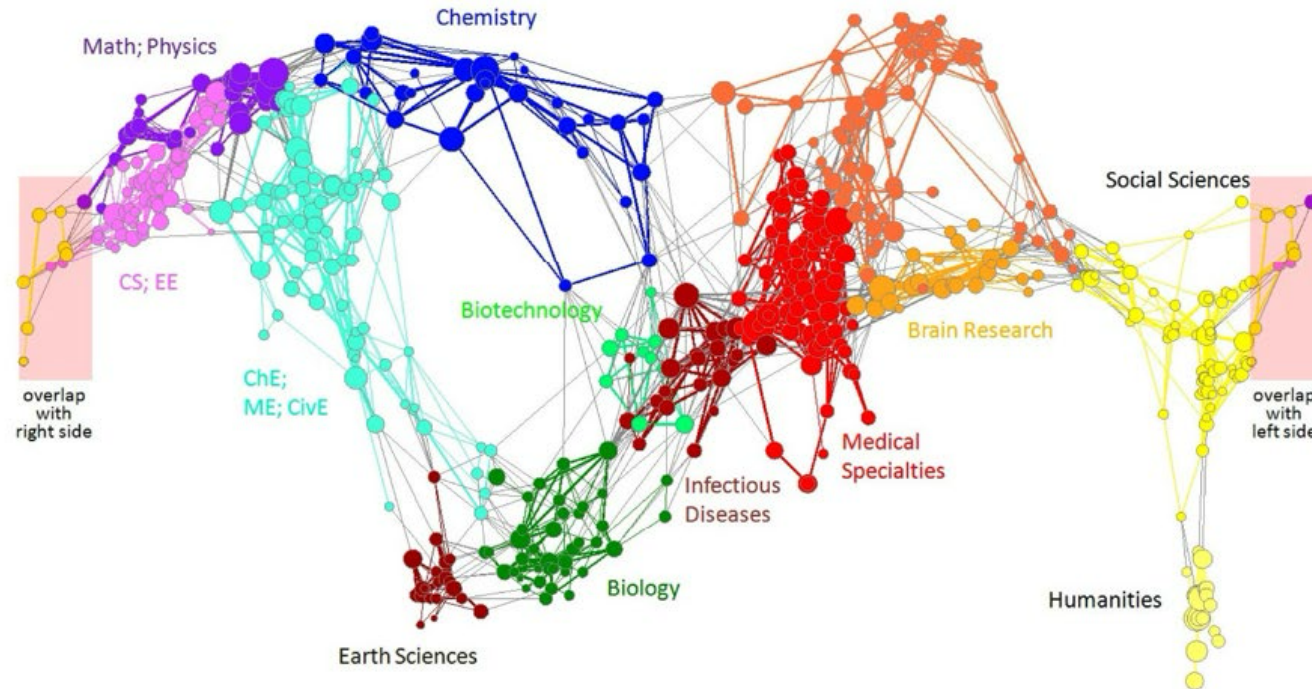
300K nodes

1.6M facts

Wikipedia Graph: <https://wiki-insights.epfl.ch/wikitrends/>

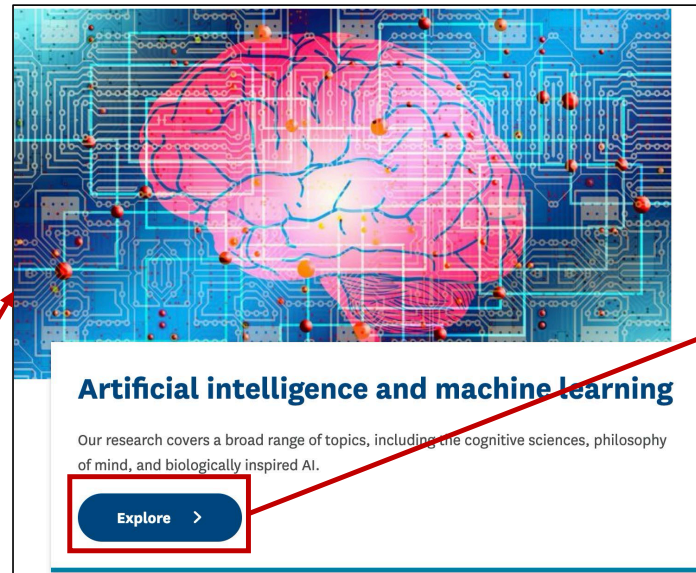
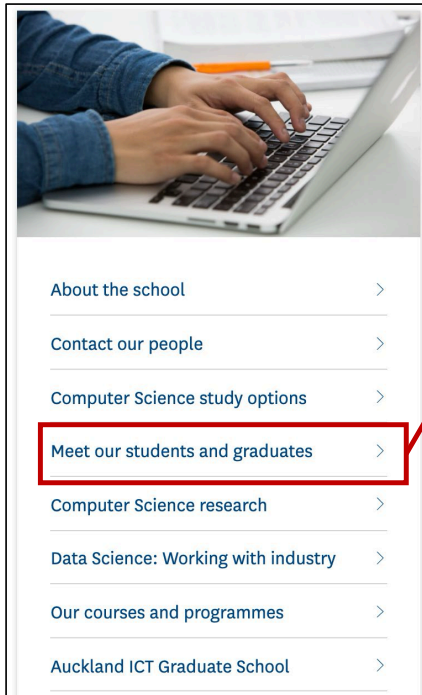
Graph data

- Citation network (Börner et al., 2012)



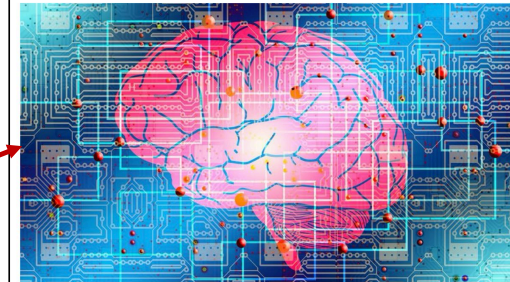
Graph data

- Web as a graph (1.13 billion websites by May 2023, according to PCWorld magazine)
- Nodes: Webpages, Edges: Hyperlinks



Artificial intelligence and machine learning

This broad research area covers interdisciplinary collaboration on topics such as the cognitive sciences, philosophy of mind, and biologically inspired AI.



Related links

- School of Computer Science >
- Computer Science research >
- Take 10 with... Mark Wilson >
- High-powered computer sees red >

Visit the Machine Learning Group website >

Artificial intelligence is the study and design of a system that perceives its environment and takes actions that maximize its chances of success.

Overview

- Web Search
 - PageRank
 - Web spam
- Social network analysis
 - Community detection
 - Influence maximization

Web search

- A broad question: How to efficiently organize the data on the web?

- **Web directories**, e.g., Yahoo!



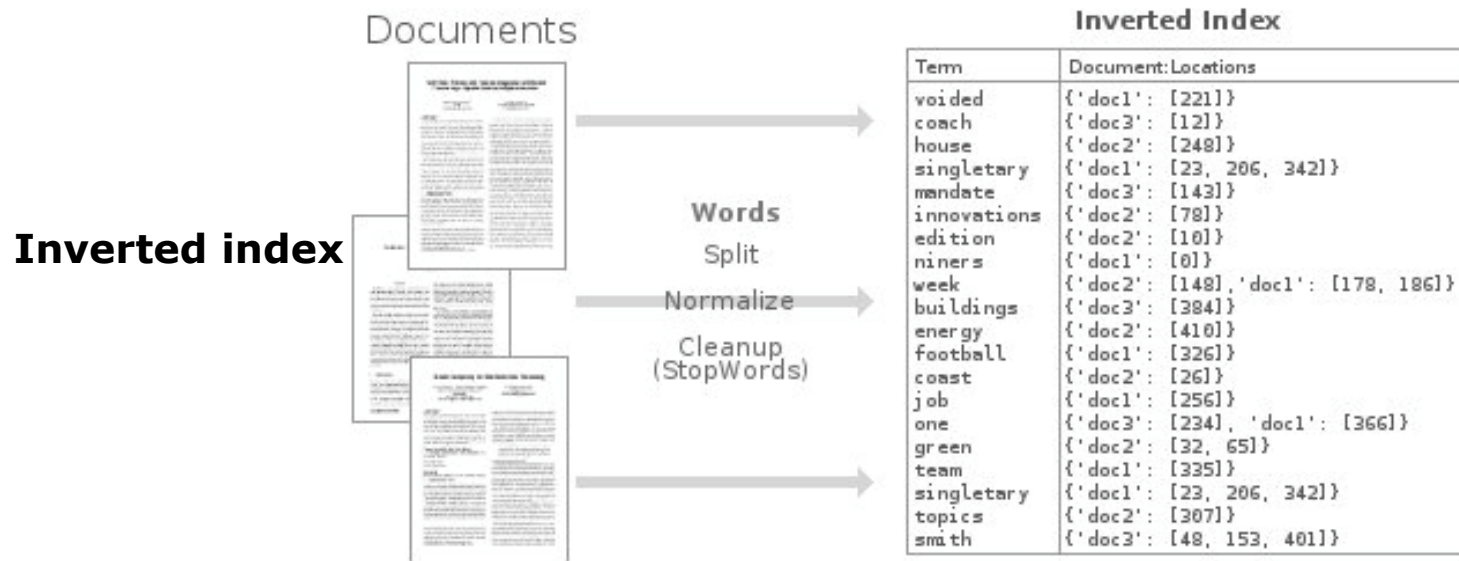
- **Web search**, e.g., Google search
 - Find a ranked list of relevant documents given keywords
 - Difficulties:
 - **Efficiency issue**: scanning over all the documents each time is impossible!
 - **Accuracy issue**: we have spam everywhere over the web!



Web search

■ Search engines:

- Crawl the Web and collect information from webpages
- *Index* webpages according to words – *inverted index*
- Answer *queries* (lists of words) with the webpages containing those words



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- Crawl the Web and collect information from webpages
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- **Search Result Ranking:**

- Return a list of webpages to a query ordered by *importance/relevance*

Importance measure by TF-IDF

- We may consider each web page as a plain text document and use *Term Frequency* - *Inverse Document Frequency* (TF-IDF)
- It models the importance of each word in a document:

$$TF - IDF(w, d) = tf(w, d) \times idf(w)$$

$$tf(w, d) = f_{t,d}, \quad idf(w) = \log_2 \frac{N}{|\{d \in D: w \in d\}|}$$

Term frequency in document

#docs contain the term

Problems with TF-IDF

- The search engine can be easily fooled
- **Term spam:**
 - Put invisible words in the webpages
 - A webpage about pizza could pretend as a page about AI, electronic, etc.:
 - Put 1000 keywords like “graph” and “algorithm” in the webpage in an invisible div layer. Or simply make the font color the same as background color.
 - Human cannot see such keywords, but the search engine does
- Idea – **see what others talk about the webpage rather than what it says about itself.**

Web as a directed graph

- **In-links** of a webpage p :
hyperlinks towards p
- **Out-links** of a webpage p :
links within p

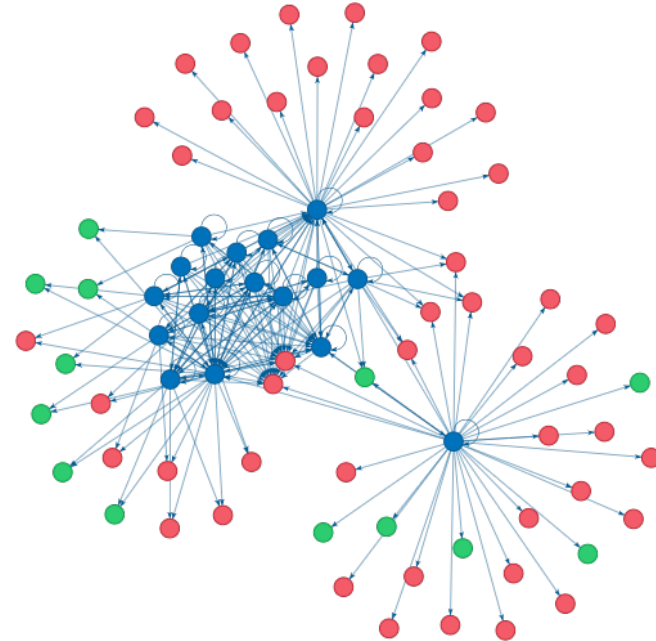


Image from:

<https://www.cs.cornell.edu/~kt/post/site-graph/>

PageRank

[Larry Page, 99']

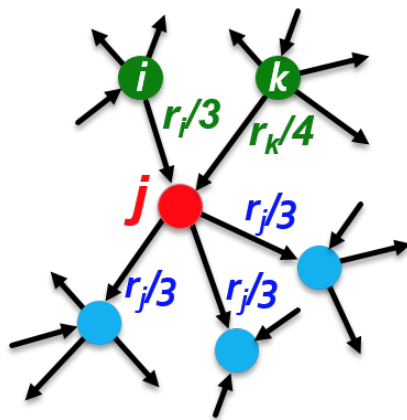
- Idea – the more *in-links* comes from important webpages, the more likely the webpage itself is important
- Page is more important if it has more in-links.
 - www.stanford.edu has 23,400 in-links
 - www.joe-schmoe.com has 1 in-link
- Links from important pages count more.
 - Many trustworthy sites like BBC news link to www.stanford.edu

PageRank

Flow definition

- Each link's vote is proportional to the *importance* of its source page
- If page j with importance r_j has d_j out-links, each link gets $\frac{r_j}{d_j}$ votes
- Page j 's importance is the sum of the votes from its in-links $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$

$$r_j = r_i/3 + r_k/4$$



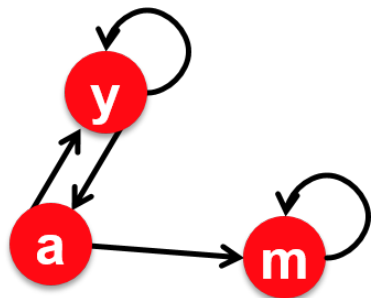
To compute the rank of each node, we need to solve a system of N linear equations!

PageRank

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- Example: Find the page rank of the following graph



$$\begin{aligned} r_a &= \frac{1}{2} r_y \\ r_y &= \frac{1}{2} r_a + \frac{1}{2} r_y \\ r_m &= r_m + \frac{1}{2} r_a \end{aligned}$$



$$\begin{aligned} r_a &= 0 \\ r_y &= 0 \\ r_m &= 1 \end{aligned}$$

$$r_a + r_y + r_m = 1$$

In this example $N = 3$, how if $N = 1$ million?

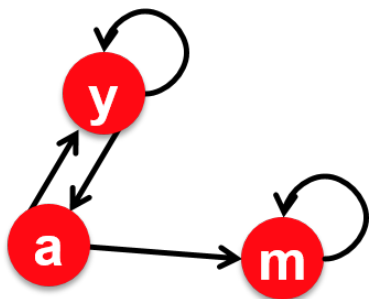
PageRank

Matrix formulation

- Let G be a graph with N nodes and $N \times N$ adjacency matrix $\mathbf{M}$
- For each node i with d_i out-links, if we have link $i \rightarrow j$, then $\mathbf{M}_{ji} = \frac{1}{d_i}$, otherwise $\mathbf{M}_{ji} = 0$
- PageRank computes the rank vector $\mathbf{r}$ as:

$$\mathbf{r} = \mathbf{M} \cdot \mathbf{r}$$

- Example of adjacency matrix (rows denote destinations, while columns denote sources):



	y	a	m
y	$\frac{1}{2}$	$\frac{1}{2}$	0
a	$\frac{1}{2}$	0	0
m	0	$\frac{1}{2}$	1

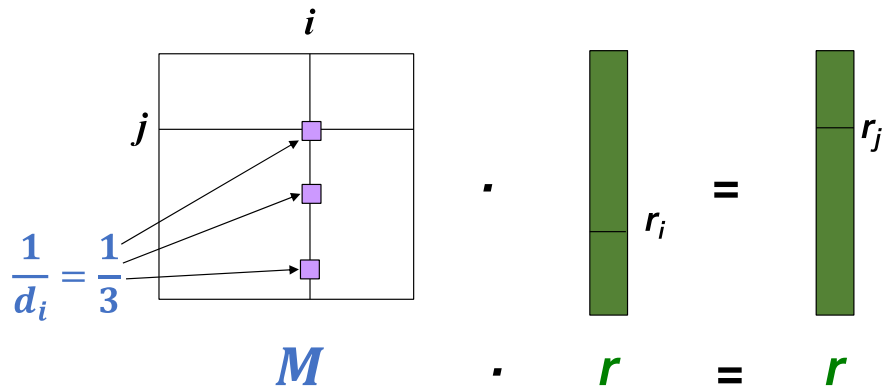
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$$r = M \cdot r$$

Relation to recursive definition $r_j = \sum_{i \rightarrow j} \frac{r_i}{d_i}$



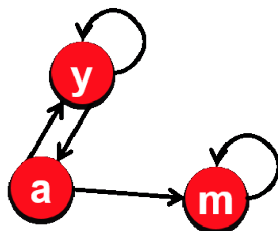
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- Suppose $\mathbf{M}$ is a **column stochastic** matrix:
 - Each entry is *non-negative*
 - Each column *sums up to 1*



	y	a	m
y	1/2	1/2	0
a	1/2	0	0
m	0	1/2	1

PageRank

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- Properties of column stochastic matrix M :
 - Always has an eigen value equals 1
 - For all other eigen values λ' , $|\lambda'| \leq 1$

NOTE: x is an eigenvector with the corresponding eigenvalue λ if:

$$Ax = \lambda x$$

For $\lambda = 1$, $MMr = Mr = r$

More information:

http://people.math.harvard.edu/~knill/teaching/math19b_2011/handouts/lecture33.pdf

Solving PageRank – Power Iteration

$$\mathbf{r} = \mathbf{M} \cdot \mathbf{r}$$

- Consider the rank vector as the eigen vector of $\mathbf{M}$ with eigen value equals 1, i.e., the stable equilibrium distribution of $\mathbf{M}$
- **Power iteration:**
 - Suppose there are N webpages
 - Initialize: $\mathbf{r}^{(0)} = [1/N, \dots, 1/N]^T$
 - Iterate: $\mathbf{r}^{(t+1)} = \mathbf{M} \cdot \mathbf{r}^{(t)}$
 - Stop when the difference of two vectors in consecutive iterations is smaller than a threshold: $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \epsilon$

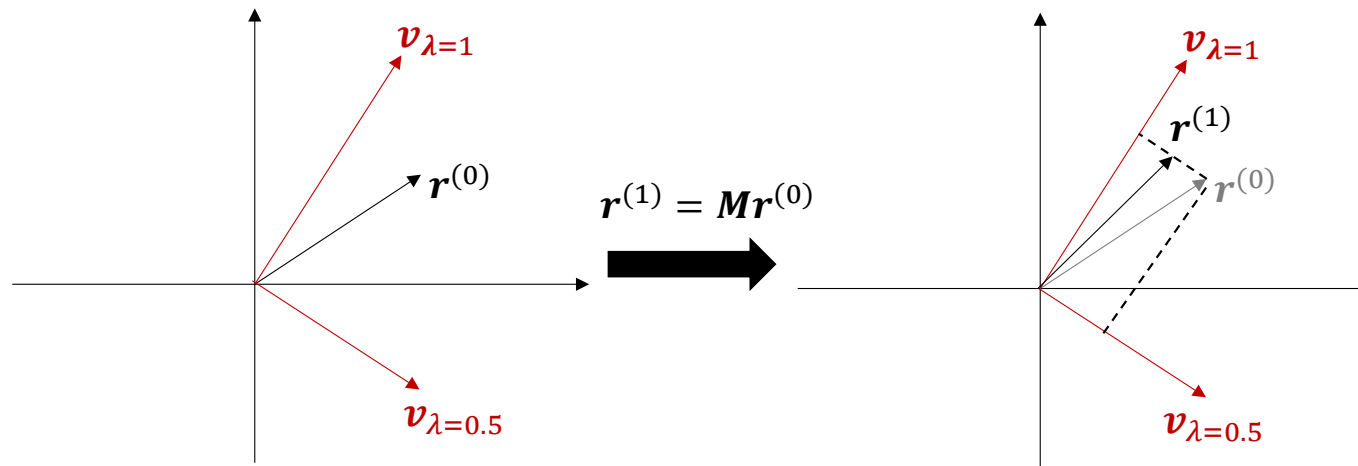
Why power iteration can converge?

Demo: <https://colab.research.google.com/drive/12SMZGqHz1NoiVq8h-Z-36sRDXwl358gg?usp=sharing>

Convergence of Page Rank - Intuition

- Remind that the largest eigen value of column stochastic matrix M is 1.

Example: Let $v_{\lambda=1}$ and $v_{\lambda=0.5}$ be the eigen vectors of column stochastic M for eigen values $\lambda = 1$ and $\lambda = 0.5$, respectively



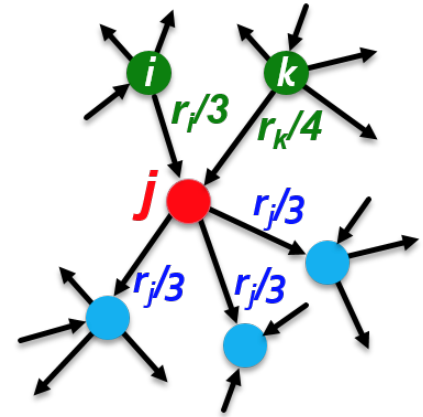
Iteratively left multiplies with $M \rightarrow$ the rank vector becomes the eigen vector $v_{\lambda=1}$ of M !

Convergence of Page Rank – Theory (Not examinable)

- A random web surfer (random walker)

1. The surfer starts at a random page i .
2. Next time, the surfer will visit any of its out-links uniformly at random
3. Suppose we select a link to page j
4. Repeat 2 and 3 from page j

$$r_j = r_i/3 + r_k/4$$



- Page probability

- Let $p(x_t)$ denotes the probability distribution over pages at time t .
- The Page Rank vector $\mathbf{r}$ is the stationary distribution $p(x_t)$ of the random walk.

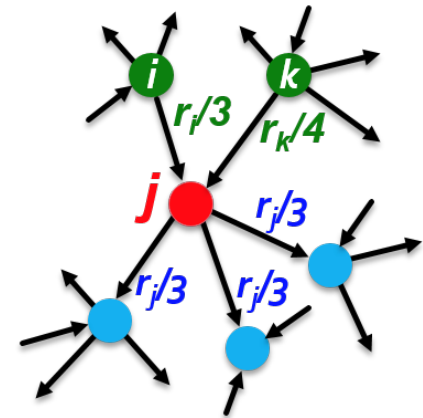
Markov Chain

- Let x_t be the page of the random walker at time t .

$$p(x_t = j) = \sum_i p(x_t = j | x_{t-1} = i) p(x_{t-1} = i)$$

- This probabilistic model is called a **Markov Chain**

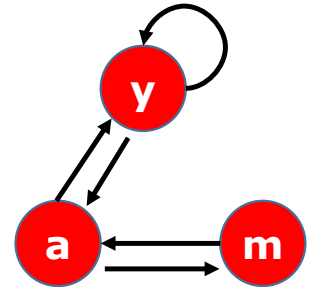
$$r_j = r_i/3 + r_k/4$$



Convergence of Random Walk

Theorem for Markov Chain:

- If the matrix M is
 - Stochastic – entries are non-negative and each column sums to 1
 - Aperiodic – when starting from a node, we don't know when we will return to the same node after transitions
 - Irreducible – all nodes can reach to any other nodes
- Then
 - The stationary distribution exists and is unique, and it does not rely on the initial values!



	y	a	m
y	$\frac{1}{2}$	$\frac{1}{2}$	0
a	$\frac{1}{2}$	0	1
m	0	$\frac{1}{2}$	0